Active Maintenance Report: coppock_hill_vavreck_2020

Coppock, Hill and Vavreck (2020, Science Advances): The Small Effects of Political Advertising are Small Regardless of Context, Message, Sender, or Receiver

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Archived at Zenodo: doi:10.5281/zenodo.24168360. This DOI always resolves to the latest release of the maintained version.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Coppock, Hill and Vavreck (2020), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1126/sciadv.abc4046
Replication archive	10.7910/DVN/TN7KWR
Pre-analysis plan	None

The data are not redistributed here. The deposit is 9.5 MB across 21 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the hand-reviewed extraction of every numeric token in the article and its supplementary materials, and the comparison table is built against it. `errata.qmd` at the root renders `coppock_hill_vavreck_2020_errata.pdf`, a note listing the sentences in the article whose numbers are wrong. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_hill_vavreck_2020.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 21 files, produces every table and figure into `maintained/output/`, and rebuilds the ground truth table from those outputs. Required packages: tidyverse, estimatr, metafor, modelsummary, janitor, broom, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about ten seconds once the deposit is on disk, most of it in the four estimation scripts, which fit more than a thousand weighted regressions across weeks and partisanship-by-battleground subgroups; the first run also fetches 9.5 MB from Dataverse. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Not any more, and the failure is silent about how much it costs. The four `estimate_*.R` scripts that build every intermediate estimate file end in `write_rds(x, path = "...")`. The `path` argument was soft-deprecated in readr 1.4 and removed in readr 2.2, so all four now stop with `argument "file" is missing, with no default` after the regressions have run. Changing `path =` to `file =` is the whole fix, and with it every script in the deposit runs to completion and reproduces its own deposited output object exactly.

What makes the failure quiet is that the eight downstream scripts do not notice. The archive deposits the four `.rds` estimate files alongside the code that makes them, so `table_1.R`, `figure_1.R`, `figure_2.R`, `table_S2.R` and the rest read the deposited objects and run clean whether or not the estimation step succeeded. An archive that ships its own intermediates can look healthy while the half of it that does the work has stopped executing.

Reproducing is a different question from running, and here the archive does reproduce. Of the 342 published quantities recorded in the ground truth, 326 are reproduced exactly by the deposited code once the four `path =` calls are repaired. The 12 that do not match are not failures of the code: they are places where the article or its supplementary materials state something the code does not produce, listed under the errata section below. A further 4 are claims about shape whose wording sets no threshold, so no verdict is computed for them.

1.2 Does the maintained rewrite reproduce the paper?

Yes. All 326 reproducible quantities match the published values to reported precision, and the 12 that do not are the same published errata the archive fails on. The rewrite also agrees with the deposit itself. Its four per-week estimate files are the objects the archive deposits

alongside its code, and they are the input to every published table and figure; across all 651 estimates in them the largest disagreement is 4.6×10^{-15}, which is floating-point noise. `maintained/text_archive_agreement.R` measures that and writes it out, so the claim is a number in `output/` rather than a sentence here.

The rewrite also removes the archive's one genuinely fragile dependency. The published pooling runs through `rmeta`, which has not been updated since March 2018 and which the archive props up with hand-written `tidy()` and `glance()` methods in its own `helpers.R`. The rewrite pools with `metafor` instead, which supports `broom` natively, so both the package and the glue holding it together are gone. No pooled estimate moves.

2 Paper overview

Citation: Coppock, A., Hill, S. J. and Vavreck, L. (2020). "The small effects of political advertising are small regardless of context, message, sender, or receiver: Evidence from 59 real-time randomized experiments." *Science Advances*, 6(36), eabc4046. DOI: 10.1126/sciadv.abc4046

Summary: Forty-nine real 2016 presidential campaign advertisements were tested in 59 advertisement-by-week experiments on 34,000 YouGov respondents across 29 weeks, from the primaries through election day. Each week a fresh national sample was assigned at random to watch one of that week's advertisements, both of them, or a placebo advertisement for car insurance, and then answered a short survey. Treatment effects are estimated week by week with covariate-adjusted weighted least squares and robust standard errors, then pooled by random-effects meta-analysis. The average effect on target candidate favorability is 0.049 scale points on a five-point scale and the average effect on vote intention is 0.7 percentage points. The paper's argument is about heterogeneity rather than about the averages: regressing the conditional effects on message, sender, receiver and context moderators (Table 1) turns up almost nothing, and the estimated standard deviation of true effects is 0.07 favorability points and 2 percentage points of vote intention. Small average effects are not concealing large offsetting ones.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
helpers.R	Sourced by five scripts; defines tidy() and glance() methods for rmeta objects	No change required
estimate_favorability_sates.R	Error: write_rds(path =) removed in readr 2.2	write_rds(x, file = ...)
estimate_favorability_cates.R	Error: write_rds(path =) removed in readr 2.2	write_rds(x, file = ...)

estimate_vote_choice_sates.R	Error: write_rds(path =) removed in readr 2.2	write_rds(x, file = ...)
estimate_vote_choice_cates.R	Error: write_rds(path =) removed in readr 2.2	write_rds(x, file = ...)
table_1.R	Clean; needs texreg, which installs from CRAN	install.packages('texreg')
figure_1.R	Clean; drops two placeholder rows, as intended	No change required
figure_2.R	Clean	No change required
in_text_numbers.R	Clean	No change required
table_S1.R	Clean; deprecation notice from spread()	No change required
table_S2.R	Clean; deprecation notice from do()	No change required
table_S3.R	Clean	No change required
figure_S1.R	Clean; deprecation notice from do()	No change required

Every package the archive names still installs from CRAN: `tidyverse`, `estimatr`, `texreg`, `xtable`, `rmeta`, `metafor` and `janitor`. `texreg` is the only one not already present in a standard installation, and version 1.39.5 installs without incident; the rewrite no longer needs it. `rmeta` is the fragile one, and the meta-analysis section below deals with it.

The deprecation notices are not errors. `do()` still works in dplyr 1.2.1, as does `spread()` in tidyr 1.3.2; both are superseded rather than removed, and the rewrite replaces them anyway.

One structural feature of the deposit is worth naming, because it shapes what can be checked. The only files any script writes are the four `.rds` estimate objects. Every table goes to the console as LaTeX through `print.xtable` or `texreg`, and every figure is printed to the default graphics device, which is why running the deposit leaves an `Rplots.pdf` holding whichever figure was drawn last. There is no deposited table or figure file to compare a re-run against, so the reproduction check has to run through the published document. The rewrite writes every table and every figure to `maintained/output/`, alongside a CSV of the values each figure plots, for that reason.

3.1 The archive overwrites its own deposit

`estimate_favorability_sates.R` and its three siblings write `favorability_sates.rds`, `favorability_cates.rds`, `vote_choice_sates.rds` and `vote_choice_cates.rds` into the working directory. Those four filenames are themselves deposited files. Running the archive in place therefore overwrites four of its own 21 deposited objects, and running `figure_1.R` or `figure_2.R` in place leaves an `Rplots.pdf` that was never part of the deposit.

Here that overwrite was harmless: the regenerated objects are value-identical to the deposited ones, which is the check that says the archive is internally reproducible. It is harmless only because the deposit was verified against Dataverse afterwards. The general form of the hazard is that a deposit can be silently replaced by the output of the code it contains, and `original/` in this repository is checked back against the served bytes before every commit for that reason.

3.2 Checksums

Table 2: Deposit verification, 1 August 2026.

Quantity	Value
Files in the deposit	21
Total size	9.5 MB
md5_served matches the local copy	21 of 21
md5_published disagrees with md5_served	0
Files Dataverse ingested as tabular	1

All 21 published checksums agree with the bytes `?format=original` returns, which is not something to assume: another deposit in this program carries three published checksums that verify neither the original file nor the tabular file derived from it. One file, `ad_df.csv`, was ingested into a `.tab` representation, so the manifest records `originalFileName` and `originalFileSize` rather than the names and sizes of the derived file.

4 Errata in the published article

12 recorded quantities do not match. None of them is a defect in the code: in every case the deposited script and the maintained rewrite agree with each other, and the article or its supplementary materials state something else. That is what `defect_locus` records in the ground truth, and every failing row carries the same value.

11 of the 12 are sentences a reader would quote, and those are collected in `coppock_hill_vavreck_2020_errata.p` at the root of this repository, rendered from `errata.qmd` with every corrected value computed from the deposit at render time. The remaining one is a table cell rather than a sentence and is described below.

Location	Quantity	Published	Code	Locus	Note
Materials and Methods	Treatment group manipulation check, minimum (%)	92	91.420000	paper_internal	The article
Materials and Methods	Treatment group manipulation check, maximum (%)	95	95.540000	paper_internal	The article
Results	Largest significant positive favorability estimate in October	0.5	0.432000	paper_internal	The article
Results	Partisanship cells of Table 1 whose standard error exceeds the coefficient		6.000000	paper_internal	The pipelin
Results	General election coefficient (scale points)	0.121	0.122600	paper_internal	The article

Location	Quantity	Published	Code	Locus	Note
Results	CATE vote choice homogeneity p-value	0.96	0.096100	paper_internal	The article
Table 1	Table 1, model 1, battleground_d, estimate	-0.00	-0.007681	paper_internal	The article
Appendix A	Distinct campaign advertisements	50	49.000000	paper_internal	The article
Appendix A	Advertisements fielded in more than one week	8	7.000000	paper_internal	The article
Appendix A	Anti-Trump deployments	28	30.000000	paper_internal	The article
Appendix A	Anti-Clinton deployments	9	11.000000	paper_internal	The article
Appendix A	Pro-Clinton deployments	9	8.000000	paper_internal	The article

They are set out as 9 numbered entries in the note. The ones worth stating in prose here carry the note's numbering.

Entry 1, the vote choice CATE homogeneity test. The Results section reports "For the set of conditional average treatment effects (CATEs), P values are 0.0002 and 0.96." Table S2 of the same paper reports 0.0961 for the vote choice CATEs, which is what the code produces. The main text figure is off by a factor of ten, and it is the one that reads as decisive evidence of homogeneity.

Table 1's battleground coefficient, which the note folds into entry 2. The published table prints `-0.00` for the battleground coefficient in the first favorability model. The code gives −0.008, and the Results section on the facing column quotes the magnitude correctly as 0.008. The table cell lost a digit in typesetting.

Entry 2, the general election coefficient. The Results section says general election advertisements move favorability "by 0.121 scale points more than primary advertisements." Table 1 prints 0.123, and 0.123 is what the code gives. The confidence bound quoted in the same sentence, "about 0.25," follows from 0.123 rather than from 0.121.

Entries 4, 5, and 8, the appendix's advertisement counts. Appendix section A says Table S1 shows "50 unique advertisements" and that, "including the ads we deployed more than once," the study tested 28 anti-Trump, nine anti-Clinton, nine pro-Clinton and three pro-Trump advertisements. The data hold 49 distinct campaign advertisements and one placebo, and the main text's 49 is right. The four counts match neither the distinct-advertisement counts (24, 11, 6, 3) nor the deployment counts (30, 11, 8, 3); the main text's own breakdown, 30/11/8/3, is the deployment count and is correct. Appendix section A also names eight advertisements as having run in multiple weeks; seven did, and "Man of People" ran once, on 14 March. Section B and Figure S1 both use the correct seven.

Entry 3, the largest estimate in the study. The Results section asks the reader to imagine what a significance filter would leave, and describes "one large positive result toward the end of the campaign in October (more than +0.5 points)." No estimate anywhere in the study reaches half a scale point. The largest is 0.432, on the advertisement "Example" in the week of

31 October, and it is in the right place on the right date. The passage's point survives; the number does not.

Entry 7, standard errors and the partisanship coefficients. The Results section says the differences in treatment response across Democrats, Independents and Republicans are "small, with SEs greater than coefficients." That holds in 6 of the 8 partisanship cells in Table 1. The exceptions are the Democratic respondent coefficients in columns 1 and 3, and the second is visible on the published page: 0.011 with a standard error of 0.010. Neither is distinguishable from zero, so the claim the sentence supports is unaffected.

Entry 6, the manipulation check range, is a smaller case. The article says treated respondents claiming to have seen a campaign advertisement ranged from 92 to 95 percent. The unweighted range across the ten waves in which the question was asked is 91.4 to 95.5 percent, and weighting by the survey weights widens it to 90.2 to 95.9. The control range, stated as 2 to 4 percent, is right.

5 Number-by-number comparison

Table 4: Ground truth: published value against the value the deposited scripts produce on current R, once the four deprecated write calls are repaired.

Location	Quantity	Paper	Archive	Match
Title	Advertisement-week experiments	59	59	1
Abstract	Distinct campaign advertisements tested	49	49	1
Abstract	Advertisement-week experiments	59	59	1
Abstract	Respondents completing a survey	34000	34000	1
Introduction	Months spanned by the fielding period	8	7.81930184804928	1
Introduction	Distinct campaign advertisements tested	49	49	1
Introduction	Distinct campaign advertisements, restated	49	49	1
Introduction	Pooled favorability effect (scale points)	0.05	0.0492278297226765	1
Introduction	SD of favorability effects across experiments (scale points)	0.07	0.0681856141500513	1
Introduction	Pooled vote choice effect (percentage points)	0.7	0.718114897405216	1
Introduction	SD of vote choice effects (percentage points)	2	2.2186653517172	1
Introduction	Weekly estimates exclude large persuasive effects			
Materials and Methods	Weeks of fielding	29	29	1
Materials and Methods	Smallest weekly sample	1000	1000	1
Materials and Methods	Largest weekly sample	2000	2000	1
Materials and Methods	Share starting but not finishing (%)	42	42.4450688966381	1
Materials and Methods	Attrition tests conducted	29	29	1
Materials and Methods	Attrition tests significant unadjusted	3	3	1
Materials and Methods	Attrition tests significant after Holm	0	0	1
Materials and Methods	Attrition tests significant after Benjamini-Hochberg	0	0	1
Materials and Methods	Anti-Trump advertisement deployments	30	30	1
Materials and Methods	Anti-Clinton advertisement deployments	11	11	1
Materials and Methods	Pro-Clinton advertisement deployments	8	8	1
Materials and Methods	Pro-Trump advertisement deployments	3	3	1
Materials and Methods	Control group manipulation check, minimum (%)	2	1.5625	1
Materials and Methods	Control group manipulation check, maximum (%)	4	4.01459854014598	1
Materials and Methods	Treatment group manipulation check, minimum (%)	92	91.4156626506024	0
Materials and Methods	Treatment group manipulation check, maximum (%)	95	95.5431754874652	0
Materials and Methods	Respondents in a single weekly study	1000	1000	1

Table 4: Ground truth: published value against the value the deposited scripts produce on current R, once the four deprecated write calls are repaired. *(continued)*

Location	Quantity	Paper	Archive	Match
Results	Experimental comparisons plotted in Figure 1	59	59	1
Results	Pooled favorability effect (scale points)	0.049	0.0492278297226765	1
Results	The pooled favorability effect is statistically significant		1	1
Results	Pooled vote choice effect (percentage points)	0.7	0.718114897405216	1
Results	The pooled vote choice effect is not statistically significant		1	1
Results	Favorability SATE homogeneity p-value	0.09	0.0919044204718359	1
Results	Vote choice SATE homogeneity p-value	0.07	0.0699520982406853	1
Results	Largest significant negative favorability estimate in May	-0.5	-0.520943603069464	1
Results	Largest significant positive favorability estimate in October	0.5	0.43202600389018	0
Results	Other significant negative favorability estimates in October	2	2	1
Results	Smallest weekly sample	1000	1000	1
Results	Largest weekly sample	2000	2000	1
Results	Partisanship cells of Table 1 whose standard error exceeds the coefficient		6	0
Results	No partisanship coefficient in Table 1 is distinguishable from zero		0	1
Results	Specifications in which the time slope is negative	3	3	1
Results	Specifications in which the time slope is distinguishable from zero	1	1	1
Results	Battleground difference in favorability, magnitude	0.008	0.0076808718197986	1
Results	General election coefficient (scale points)	0.121	0.122624298396105	0
Results	The general election coefficient is not statistically significant		1	1
Results	Upper bound of the general election interval	0.25	0.253750226781251	1
Results	No tone or sponsor coefficient in Table 1 is distinguishable from zero		0	1
Results	Target candidate coefficients relative to the omitted pro-Clinton category		6	
Results	No target candidate coefficient in Table 1 is distinguishable from zero		0	1
Results	Democratic respondents respond more strongly to pro-Democratic than pro-Republican advertisements		1	1
Results	Republican respondents respond similarly to pro-Democratic and pro-Republican advertisements			
Results	SATE favorability homogeneity p-value, restated	0.09	0.0919044204718359	1
Results	SATE vote choice homogeneity p-value, restated	0.07	0.0699520982406853	1
Results	CATE favorability homogeneity p-value	0.0002	0.000165056906355207	1
Results	CATE vote choice homogeneity p-value	0.96	0.0961039371758417	0
Results	Homogeneity tests in Table S2 that fail to reject at the 5 per cent level	3	3	1
Results	Tau, favorability SATEs	0.07	0.0681856141500513	1
Results	Tau, favorability CATEs	0.15	0.147422967808586	1
Results	Tau, vote choice SATEs	0.02	0.022186653517172	1
Results	Tau, vote choice CATEs	0.02	0.0232088109488414	1
Discussion	Advertisement-week experiments	59	59	1
Discussion	Respondents completing a survey	34000	34000	1
Discussion	Distinct campaign advertisements tested	49	49	1
Figure 1	Favorability meta-analytic estimate	0.049	0.0492278297226765	1
Figure 1	Favorability meta-analytic standard error	0.020	0.0200009449806686	1
Figure 1	Vote choice meta-analytic estimate	0.007	0.00718114897405216	1
Figure 1	Vote choice meta-analytic standard error	0.007	0.00732724536941071	1
Table 1	Table 1, model 1, intrcpt, estimate	0.056	0.0559513746296691	1
Table 1	Table 1, model 1, intrcpt, std.error	0.020	0.0203091094511682	1
Table 1	Table 1, model 2, intrcpt, estimate	0.062	0.0619557871802225	1
Table 1	Table 1, model 2, intrcpt, std.error	0.020	0.0204550273103208	1
Table 1	Table 1, model 3, intrcpt, estimate	0.007	0.0071734391731233	1
Table 1	Table 1, model 3, intrcpt, std.error	0.007	0.0071442025839593	1
Table 1	Table 1, model 4, intrcpt, estimate	0.008	0.00761340879292042	1

Table 4: Ground truth: published value against the value the deposited scripts produce on current R, once the four deprecated write calls are repaired. *(continued)*

Location	Quantity	Paper	Archive	Match
Table 1	Table 1, model 4, intrcpt, std.error	0.007	0.00721776470356415	1
Table 1	Table 1, model 1, democrat_d, estimate	0.035	0.035239216310936	1
Table 1	Table 1, model 1, democrat_d, std.error	0.035	0.0346822419625252	1
Table 1	Table 1, model 2, democrat_d, estimate	0.022	0.0217616219401439	1
Table 1	Table 1, model 2, democrat_d, std.error	0.036	0.0361069235707198	1
Table 1	Table 1, model 3, democrat_d, estimate	0.011	0.010806619726571	1
Table 1	Table 1, model 3, democrat_d, std.error	0.010	0.010032366314141	1
Table 1	Table 1, model 4, democrat_d, estimate	0.006	0.00607624424823846	1
Table 1	Table 1, model 4, democrat_d, std.error	0.011	0.0111901459055996	1
Table 1	Table 1, model 1, independent_d, estimate	0.023	0.0230994168403524	1
Table 1	Table 1, model 1, independent_d, std.error	0.051	0.0514908172832447	1
Table 1	Table 1, model 2, independent_d, estimate	0.015	0.0148795264572918	1
Table 1	Table 1, model 2, independent_d, std.error	0.052	0.0518214653308673	1
Table 1	Table 1, model 3, independent_d, estimate	0.009	0.00882756839601492	1
Table 1	Table 1, model 3, independent_d, std.error	0.020	0.0196989007331031	1
Table 1	Table 1, model 4, independent_d, estimate	0.007	0.00662959611432708	1
Table 1	Table 1, model 4, independent_d, std.error	0.020	0.0199216048048094	1
Table 1	Table 1, model 1, battleground_d, estimate	-0.00	-0.0076808718197986	0
Table 1	Table 1, model 1, battleground_d, std.error	0.033	0.0330993309997604	1
Table 1	Table 1, model 2, battleground_d, estimate	-0.007	-0.00728897411725834	1
Table 1	Table 1, model 2, battleground_d, std.error	0.033	0.0331245308366241	1
Table 1	Table 1, model 3, battleground_d, estimate	-0.017	-0.0169737781355814	1
Table 1	Table 1, model 3, battleground_d, std.error	0.010	0.00982144227142084	1
Table 1	Table 1, model 4, battleground_d, estimate	-0.017	-0.0170054042184294	1
Table 1	Table 1, model 4, battleground_d, std.error	0.010	0.00996790251931272	1
Table 1	Table 1, model 1, pac_d, estimate	-0.012	-0.0124947275778723	1
Table 1	Table 1, model 1, pac_d, std.error	0.043	0.0426308784476082	1
Table 1	Table 1, model 2, pac_d, estimate	0.026	0.0256893089604678	1
Table 1	Table 1, model 2, pac_d, std.error	0.047	0.046985181519741 9	1
Table 1	Table 1, model 3, pac_d, estimate	-0.023	-0.0231494095931858	1
Table 1	Table 1, model 3, pac_d, std.error	0.013	0.0128349559721876	1
Table 1	Table 1, model 4, pac_d, estimate	-0.016	-0.0156649998403321	1
Table 1	Table 1, model 4, pac_d, std.error	0.014	0.0144268706850987	1
Table 1	Table 1, model 1, date_d, estimate	-0.023	-0.0230094166284181	1
Table 1	Table 1, model 1, date_d, std.error	0.014	0.0140563401835011	1
Table 1	Table 1, model 2, date_d, estimate	0.005	0.0045562321145179	1
Table 1	Table 1, model 2, date_d, std.error	0.010	0.0101010468834039	1
Table 1	Table 1, model 3, date_d, estimate	-0.009	-0.00937763297831397	1
Table 1	Table 1, model 3, date_d, std.error	0.004	0.00379382342908776	1
Table 1	Table 1, model 4, date_d, estimate	-0.008	-0.00774541926272452	1
Table 1	Table 1, model 4, date_d, std.error	0.004	0.00426985776830753	1
Table 1	Table 1, model 1, attack_d, estimate	-0.017	-0.0171013743308074	1
Table 1	Table 1, model 1, attack_d, std.error	0.046	0.0462370739155849	1
Table 1	Table 1, model 3, attack_d, estimate	0.028	0.0279600502401837	1
Table 1	Table 1, model 3, attack_d, std.error	0.016	0.0163558547902618	1
Table 1	Table 1, model 1, general_d, estimate	0.123	0.122624298396105	1
Table 1	Table 1, model 1, general_d, std.error	0.067	0.0669022132138403	1
Table 1	Table 1, model 2, pro_trump_d, estimate	-0.124	-0.123836901513909	1
Table 1	Table 1, model 2, pro_trump_d, std.error	0.101	0.101118675428876	1
Table 1	Table 1, model 4, pro_trump_d, estimate	-0.016	-0.0157727126722738	1
Table 1	Table 1, model 4, pro_trump_d, std.error	0.034	0.0337767139387826	1
Table 1	Table 1, model 2, anti_clinton_d, estimate	-0.105	-0.105067385465675	1
Table 1	Table 1, model 2, anti_clinton_d, std.error	0.070	0.0698886090607599	1
Table 1	Table 1, model 4, anti_clinton_d, estimate	0.012	0.0115687385179081	1
Table 1	Table 1, model 4, anti_clinton_d, std.error	0.023	0.02296757258421	1
Table 1	Table 1, model 2, anti_trump_d, estimate	-0.041	-0.0413676950109048	1
Table 1	Table 1, model 2, anti_trump_d, std.error	0.058	0.0579066875305163	1
Table 1	Table 1, model 4, anti_trump_d, estimate	0.026	0.0261014333861311	1
Table 1	Table 1, model 4, anti_trump_d, std.error	0.021	0.0214091583815356	1
Table 1	Table 1, model 2, pro_sanders_d, estimate	-0.075	-0.0749244096818968	1
Table 1	Table 1, model 2, pro_sanders_d, std.error	0.089	0.0885300550768095	1
Table 1	Table 1, model 2, pro_cruz_d, estimate	0.047	0.0470694919223399	1
Table 1	Table 1, model 2, pro_cruz_d, std.error	0.116	0.116193805568345	1
Table 1	Table 1, model 2, pro_kasich_d, estimate	-0.182	-0.181777858909213	1
Table 1	Table 1, model 2, pro_kasich_d, std.error	0.145	0.14476493785976	1
Table 1	Table 1, model 1, number of observations	354	354	1

Table 4: Ground truth: published value against the value the deposited scripts produce on current R, once the four deprecated write calls are repaired. *(continued)*

Location	Quantity	Paper	Archive	Match
Table 1	Table 1, model 2, number of observations	354	354	1
Table 1	Table 1, model 3, number of observations	204	204	1
Table 1	Table 1, model 4, number of observations	204	204	1
Table 1	Table 1 marks the model 1 average effect as significant at the 5 per cent level		1	1
Table 1	Table 1 marks the model 2 average effect as significant at the 5 per cent level		1	1
Table 1	Table 1 marks the model 3 time slope as significant at the 5 per cent level		1	1
Figure 2	Figure 2, Favorability, Republican respondents, pro-Democratic ad, estimate	0.039	0.0386595396179065	1
Figure 2	Figure 2, Favorability, Republican respondents, pro-Democratic ad, std.error	0.036	0.0356581601671249	1
Figure 2	Figure 2, Favorability, Republican respondents, pro-Republican ad, estimate	0.048	0.0480196665227286	1
Figure 2	Figure 2, Favorability, Republican respondents, pro-Republican ad, std.error	0.039	0.0389230315207782	1
Figure 2	Figure 2, Favorability, Independent respondents, pro-Democratic ad, estimate	0.070	0.0699932736594406	1
Figure 2	Figure 2, Favorability, Independent respondents, pro-Democratic ad, std.error	0.068	0.0678203342309325	1
Figure 2	Figure 2, Favorability, Independent respondents, pro-Republican ad, estimate	0.012	0.0121099394831964	1
Figure 2	Figure 2, Favorability, Independent respondents, pro-Republican ad, std.error	0.087	0.0871736863408394	1
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Democratic ad, estimate	0.106	0.106476573620707	1
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Democratic ad, std.error	0.027	0.0274578187061506	1
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Republican ad, estimate	-0.026	-0.0263833309845451	1
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Republican ad, std.error	0.049	0.0494319844445353	1
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Democratic ad, estimate	0.007	0.00699092320146246	1
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Democratic ad, std.error	0.014	0.0139356630863744	1
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Republican ad, estimate	-0.002	-0.00216608514888707	1
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Republican ad, std.error	0.009	0.00912830313876472	1
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Democratic ad, estimate	0.005	0.00506265749355459	1
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Democratic ad, std.error	0.024	0.0237475886684295	1
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Republican ad, estimate	0.027	0.0271627314372247	1
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Republican ad, std.error	0.029	0.0291458503455212	1
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Democratic ad, estimate	0.027	0.0271759277742814	1
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Democratic ad, std.error	0.008	0.00801715594366605	1
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Republican ad, estimate	-0.023	-0.023396092643628	1
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Republican ad, std.error	0.015	0.0147860283788054	1
Appendix A	Distinct campaign advertisements	50	49	0
Appendix A	Advertisements fielded in more than one week	8	7	0
Appendix A	Anti-Trump deployments	28	30	0
Appendix A	Anti-Clinton deployments	9	11	0
Appendix A	Pro-Clinton deployments	9	8	0
Appendix A	Pro-Trump deployments	3	3	1
Table S1	Weeks listed	29	29	1
Table S1	Advertisement-week entries	58	58	1
Appendix B	Advertisements fielded more than once	7	7	1
Appendix B	Repeat advertisements whose effect varies significantly across weeks		0	1
Appendix B	Weekly estimates for the advertisement Quotes			
Figure S1	Panels	7	7	1

Table 4: Ground truth: published value against the value the deposited scripts produce on current R, once the four deprecated write calls are repaired. *(continued)*

Location	Quantity	Paper	Archive	Match
Figure S1	Weekly estimates plotted	17	17	1
Table S2	Table S2, sates, Favorability, estimate	0.0492	0.0492278297226765	1
Table S2	Table S2, sates, Favorability, std.error	0.0200	0.0200009449806686	1
Table S2	Table S2, sates, Favorability, p.value	0.0919	0.0919044204718359	1
Table S2	Table S2, sates, Favorability, tau2	0.0046	0.00464927797701968	1
Table S2	Table S2, sates, Favorability, tau	0.0682	0.0681856141500513	1
Table S2	Table S2, sates, Vote Choice, estimate	0.0072	0.00718114897405216	1
Table S2	Table S2, sates, Vote Choice, std.error	0.0073	0.00732724536941071	1
Table S2	Table S2, sates, Vote Choice, p.value	0.0700	0.0699520982406853	1
Table S2	Table S2, sates, Vote Choice, tau2	0.0005	0.000492247594291039	1
Table S2	Table S2, sates, Vote Choice, tau	0.0222	0.022186653517172	1
Table S2	Table S2, cates, Favorability, estimate	0.0585	0.0584522714979407	1
Table S2	Table S2, cates, Favorability, std.error	0.0178	0.0178394276247739	1
Table S2	Table S2, cates, Favorability, p.value	0.0002	0.000165056906355207	1
Table S2	Table S2, cates, Favorability, tau2	0.0217	0.0217335314374913	1
Table S2	Table S2, cates, Favorability, tau	0.1474	0.147422967808586	1
Table S2	Table S2, cates, Vote Choice, estimate	0.0083	0.00826356365738537	1
Table S2	Table S2, cates, Vote Choice, std.error	0.0051	0.00514489865140912	1
Table S2	Table S2, cates, Vote Choice, p.value	0.0961	0.0961039371758417	1
Table S2	Table S2, cates, Vote Choice, tau2	0.0005	0.000538648905659061	1
Table S2	Table S2, cates, Vote Choice, tau	0.0232	0.0232088109488414	1

The table above omits the 136 cells of Table S3, which would swamp it. All 136 reproduce exactly, and they are in the ground truth CSV.

The starting point is `ground_truth/published_claims.csv`, an extraction of every numeric token in the article and its supplementary materials, classified by hand into the ones the pipeline can reach and the ones nothing in it can move. It holds 390 entries; 342 of them are quantities the pipeline produces or claims about shape it can settle, and each of those has a row in the ground truth. The rest are scale endpoints, an IRB number, section and question numbering, and the two numbers the supplementary materials quote from outside the study, and giving one of those a row would mean typing a published number into the code.

Every published float is represented. The two tables and two figures that print coefficients are covered cell by cell, as is Table S3. The other two print none, so each is checked on what it does state: Table S1 on the shape of the advertisement list, and Figure S1 on its seven panels and seventeen plotted estimates. The table is built by `ground_truth/build_ground_truth.R`, which reads every comparison value back out of `maintained/output/` and refuses to write a failing row that carries no `defect_locus`.

5.1 Two instruments, checked against each other

`maintained/in_text_claims.R` carries one block per claim: the sentence the article prints, verbatim, and the value this pipeline gives for it in the article's own units and rounding. It reads the same files in `output/` that the ground truth reads, and does its own selection, unit conversion and rounding, so the two arrive at each number by separate paths.

`build_ground_truth.R` runs that file non-interactively, counts the claims it printed, checks the set against the extraction, and compares every printed value against the one it derived for itself. A block that errors, or that quietly stops printing, fails that check; a comment saying a claim is covered would not. A disagreement between the two is a finding rather than a coincidence, and the build stops on one.

6 Maintained rewrite

The rewrite lives in `maintained/`: 17 scripts covering four estimation steps, four published tables, three figures, the in-text quantities, the claims about shape and count, the comparison against the deposit, and the in-text claims audit. It is a translation, not a reanalysis. Every estimator, specification and sample restriction is the one the paper used, including the two places where the archive's choices are not what a rule would predict: Figure S1 scores the pro-Clinton advertisement "Real Life" against Trump favorability while scoring the pro-Clinton advertisement "Love/Kindness" against Clinton favorability, and the rewrite keeps both.

6.1 Architecture

`clean_favorability_sates.R` and its three siblings replace the archive's `estimate_*.R` scripts. Each fits the paper's covariate-adjusted weighted specification within every week, or within every partisanship-by-battleground-by-week cell, and writes a tidy estimate file to `maintained/output/`. Nothing downstream reads the archive's deposited `.rds` estimate files; they are rebuilt, which is what makes the reproduction check meaningful rather than circular.

The archive's `estimate_ad_effects()` used `do(tidy(lm_robust(...)))` over a grouped frame. The rewrite uses `reframe(tidy(lm_robust(..., data = pick(everything()))))`, which is the supported replacement. The weights column in `exps.rds` is literally named `weights`, so `lm_robust` resolves it inside the data frame and the usual non-standard-evaluation trap does not arise here.

Two helper functions survive into `helpers.R`, and no more. `estimate_ad_effects()` encapsulates the per-week regression, and `pool_effects()` encapsulates the meta-analysis. `pool_effects()` takes `method` with no default, so every call site names the estimator it is asking for rather than inheriting one. The archive's `add_parens()`, `format_num()` and `make_interval_entry()` were wrappers around a single `sprintf` call or were never called at all, and are gone; `texprep()` existed only to feed the `texreg` workaround described below and is gone with it.

`text_archive_agreement.R` runs last among the analysis scripts and does the one thing the rewrite cannot claim for itself: it reads the archive's four deposited estimate objects back off disk and compares them, row for row, against the four the rewrite has just built. `run_all.R`

then sources `ground_truth/build_ground_truth.R`, which assembles the comparison table from `output/` rather than from anything typed by hand.

6.2 Confidence bounds that lost their labels

Every effect is signed so that a positive number means the advertisement worked, which for an attack advertisement means negating the estimate. All four of the archive's estimation scripts negate the two confidence bounds in place as well:

```
estimate  = if_else(attack == 1, estimate * -1, estimate),
conf.low  = if_else(attack == 1, conf.low * -1, conf.low),
conf.high = if_else(attack == 1, conf.high * -1, conf.high)
```

Negating an interval reverses it, so after those three lines `conf.low` holds the upper end and `conf.high` the lower one, on every attack advertisement's row. It affects 497 of the 651 estimates in the four deposited files.

No published number moves. Each pooled quantity in the paper is built from the estimate and its standard error, neither of which is touched, and a line range is drawn between its two ends whichever way round they arrive, so both figures are unaffected. The cost is to a reader of the deposited estimate files, or of the plotted-values CSV this repository commits beside each figure, who would take the column names at face value. The rewrite swaps the two bounds so the names stay true, which is the only place its estimate files differ from the archive's beyond floating-point noise.

6.3 Deprecated patterns replaced

Table 5: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘write_rds(x, path =)‘	‘write_rds(x, file =)‘
‘do(tidy(...))‘	‘reframe(tidy(...))‘
‘do(with(., tidy(...)))‘	‘reframe(pool_effects(...))‘
‘rmeta::meta.summaries(method = ’random’)‘	‘metafor::rma(method = ’DL’)‘
hand-written ‘tidy.meta.summaries()‘ / ‘glance.meta.summaries()‘	(omitted: broom supports rma natively)
‘texreg‘ over dummy ‘lm‘ fits	‘modelsummary(output =)‘ on the rma fits
‘xtable‘ + ‘print.xtable‘	‘write_csv()‘ and ‘kable()‘ to ‘output/‘
‘spread()‘	‘pivot_wider()‘
‘geom_errorbar(width = 0)‘	‘geom_linerange()‘
‘split(.$wave)‘	‘split(exps_inc$wave)‘
‘expand.grid()‘	‘tibble()‘
‘cat()‘ for in-text quantities	a tibble written to ‘output/‘

6.4 Table 1 without the texreg workaround

The archive builds Table 1 by fitting each meta-regression twice: once with `metafor::rma.uni()`, which produces the numbers, and once with `lm()`, which produces an object `texreg` knows how to format. The `lm` coefficients are then overridden with the `rma` ones through `override.coef`, `override.se` and `override.pvalues`, and a helper called `texprep()` extracts the three vectors. The comment in the archive is candid about it: "These lm models are here to trick texreg."

The trick is no longer necessary. `broom` provides `tidy.rma()` and `glance.rma()`, so `modelsummary` formats a list of `rma` objects directly, and the rewrite's `table_1_meta_regression.R` passes the four fits straight to `modelsummary(output = "...tex")` with a `coef_map` and nothing else. Two of the three failure modes the workaround carried disappear with it: the possibility that a `lm` fit and its `rma` twin drift apart in specification, and the possibility that a reader takes the printed sample size or fit statistic from the `lm` rather than from the meta-regression. The result reproduces all 76 published cells of Table 1, in the published row order, including the stars on the three significant coefficients.

7 Meta-analysis: rmeta to metafor

Every pooled estimate in the paper comes from `rmeta::meta.summaries(estimate, std.error, method = "random")`. The package has not been updated since March 2018. The archive's `helpers.R` also carries hand-written `tidy.meta.summaries()` and `glance.meta.summaries()` functions, because `rmeta` predates `broom` and its objects are not tidiable; those two functions are the seam holding an unmaintained package into a tidyverse pipeline.

`rmeta`'s random-effects estimator is DerSimonian-Laird, and its heterogeneity test is Woolf's, which the appendix names explicitly. Both have exact counterparts in `metafor`: `rma(yi, sei, method = "DL")` gives the same pooled estimate, the same standard error, the same normal-theory confidence interval, the same between-study variance, and a Q test whose p-value is the same number. The rewrite therefore drops `rmeta` and both hand-written methods, and `pool_effects()` in `helpers.R` returns all seven quantities from one `rma` fit.

No estimate moves. Running both implementations on all four pooled sets, the largest absolute discrepancy across the estimate, the standard error, both confidence bounds, the heterogeneity p-value and tau-squared, 24 quantities in all, is 1.1e-16, which is floating-point noise. The check is deliberately not part of the pipeline, since running it would put `rmeta` back among the repository's dependencies, but it takes five lines and about two seconds:

```r
d <- readRDS("maintained/output/favorability_sates.rds")
m <- rmeta::meta.summaries(d$estimate, d$std.error, method = "random")
r <- metafor::rma(yi = d$estimate, sei = d$std.error, method = "DL")
c(m$summary - as.numeric(r$beta), m$se.summary - r$se,
  m$het[3] - r$QEp, m$tau2 - r$tau2, summary(m)$summci[c(1, 3)] - c(r$ci.lb, r$ci.ub))
```

A general instruction to reproduce a published pooling with `method = "FE"`, reporting REML alongside, is the wrong instruction here: it presumes an archive that pooled by fixed effects, and this one did not. Fixed effects would have moved every published number, the favorability SATE from 0.049 to 0.052 and the vote choice SATE from 0.007 to 0.009. Match the estimator the archive used, which here is `"DL"`.

Reporting REML alongside is still worth doing, because the choice of between-study variance estimator is invisible in the published table and it moves the estimates.

Table 6: Table S2 as published (DL) with the REML fit of the same data beside it. The Q test does not depend on the variance estimator, so its p-value is common to both rows.

Estimand	Outcome	Estimator	k	Estimate	SE	Q p-value	tau2	tau
sates	Favorability	DL (published)	59	0.0492	0.0200	0.0919	0.0046	0.0682
sates	Favorability	REML (added here)	59	0.0498	0.0194	0.0919	0.0035	0.0592
sates	Vote Choice	DL (published)	34	0.0072	0.0073	0.0700	0.0005	0.0222
sates	Vote Choice	REML (added here)	34	0.0071	0.0074	0.0700	0.0005	0.0225
cates	Favorability	DL (published)	354	0.0585	0.0178	0.0002	0.0217	0.1474
cates	Favorability	REML (added here)	354	0.0599	0.0159	0.0002	0.0074	0.0859
cates	Vote Choice	DL (published)	204	0.0083	0.0051	0.0961	0.0005	0.0232
cates	Vote Choice	REML (added here)	204	0.0084	0.0049	0.0961	0.0003	0.0186

REML shrinks the estimated between-study variance in three of the four sets, most sharply for the favorability CATEs, where tau-squared falls from 0.0217 to 0.0074 and the implied standard deviation of true effects from 0.15 to 0.09. The paper's headline claim is that effects are homogeneous, so an estimator that finds less heterogeneity does not threaten it; a reader should nonetheless know that the 0.15 the Discussion quotes is the larger of two defensible numbers.

8 Figure verification

Both main-text figures carry their estimates as printed annotations, and every one of the fourteen reproduces: the two on Figure 1 and all twelve on Figure 2.

Table 7: Figure annotations, transcribed from pages 3 and 5 of the article, against the maintained rewrite.

Figure	Quantity	Paper	Rewrite	Match
Figure 1	Favorability meta-analytic estimate	0.049	0.0492278	1
Figure 1	Favorability meta-analytic standard error	0.020	0.0200009	1
Figure 1	Vote choice meta-analytic estimate	0.007	0.0071811	1
Figure 1	Vote choice meta-analytic standard error	0.007	0.0073272	1
Figure 2	Figure 2, Favorability, Republican respondents, pro-Democratic ad, estimate	0.039	0.0386595	1
Figure 2	Figure 2, Favorability, Republican respondents, pro-Democratic ad, std.error	0.036	0.0356582	1
Figure 2	Figure 2, Favorability, Republican respondents, pro-Republican ad, estimate	0.048	0.0480197	1
Figure 2	Figure 2, Favorability, Republican respondents, pro-Republican ad, std.error	0.039	0.0389230	1
Figure 2	Figure 2, Favorability, Independent respondents, pro-Democratic ad, estimate	0.070	0.0699933	1
Figure 2	Figure 2, Favorability, Independent respondents, pro-Democratic ad, std.error	0.068	0.0678203	1
Figure 2	Figure 2, Favorability, Independent respondents, pro-Republican ad, estimate	0.012	0.0121099	1
Figure 2	Figure 2, Favorability, Independent respondents, pro-Republican ad, std.error	0.087	0.0871737	1
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Democratic ad, estimate	0.106	0.1064766	1
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Democratic ad, std.error	0.027	0.0274578	1
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Republican ad, estimate	-0.026	-0.0263833	1
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Republican ad, std.error	0.049	0.0494320	1
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Democratic ad, estimate	0.007	0.0069909	1
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Democratic ad, std.error	0.014	0.0139357	1
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Republican ad, estimate	-0.002	-0.0021661	1
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Republican ad, std.error	0.009	0.0091283	1
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Democratic ad, estimate	0.005	0.0050627	1
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Democratic ad, std.error	0.024	0.0237476	1
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Republican ad, estimate	0.027	0.0271627	1
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Republican ad, std.error	0.029	0.0291459	1
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Democratic ad, estimate	0.027	0.0271759	1
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Democratic ad, std.error	0.008	0.0080172	1
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Republican ad, estimate	-0.023	-0.0233961	1
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Republican ad, std.error	0.015	0.0147860	1

8.1 A layout that stopped reproducing

Figure 1's values reproduce; its layout, drawn by the archive's own code on current R, does not. The archive draws the line marking the start of the general election campaign with a bare `geom_vline(xintercept = as.Date("2016-06-10"))`. Under ggplot2 3.3.2, where the paper was made, that layer sat outside the faceted data. Under ggplot2 4.x it joins the free x scale of every panel, so the meta-analysis facet, whose data span 30 days in January 2018, is stretched to a range of nineteen months; because the facet is laid out with `space = "free_x"`, the stretched panel then takes most of the plot width and the month labels of the weekly panel collapse into each other.

The archive's own `figure_1.R` produces the same broken layout today, so this is drift in ggplot2 rather than a defect introduced in translation. The rewrite supplies the rule as a one-row data layer restricted to the weekly facet, which restores the published proportions and changes nothing that is plotted.

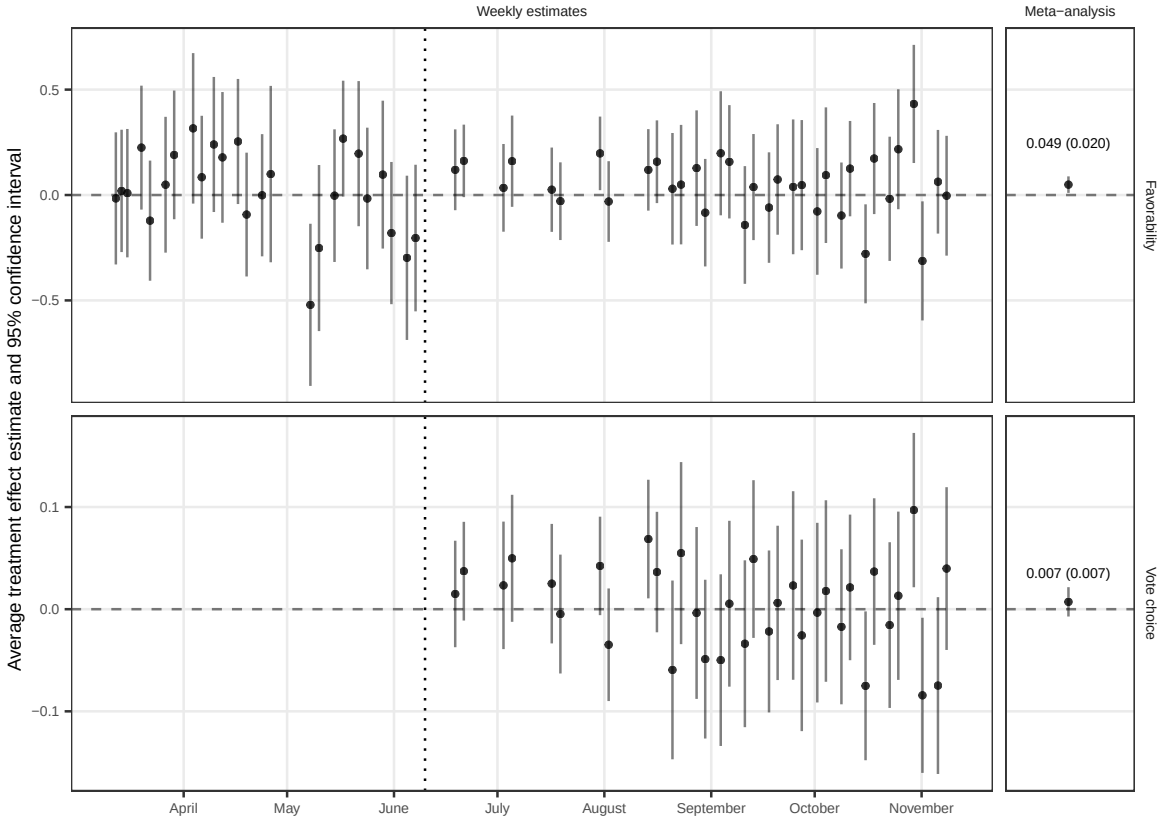


Figure 1: Figure 1 as reproduced by the maintained rewrite.

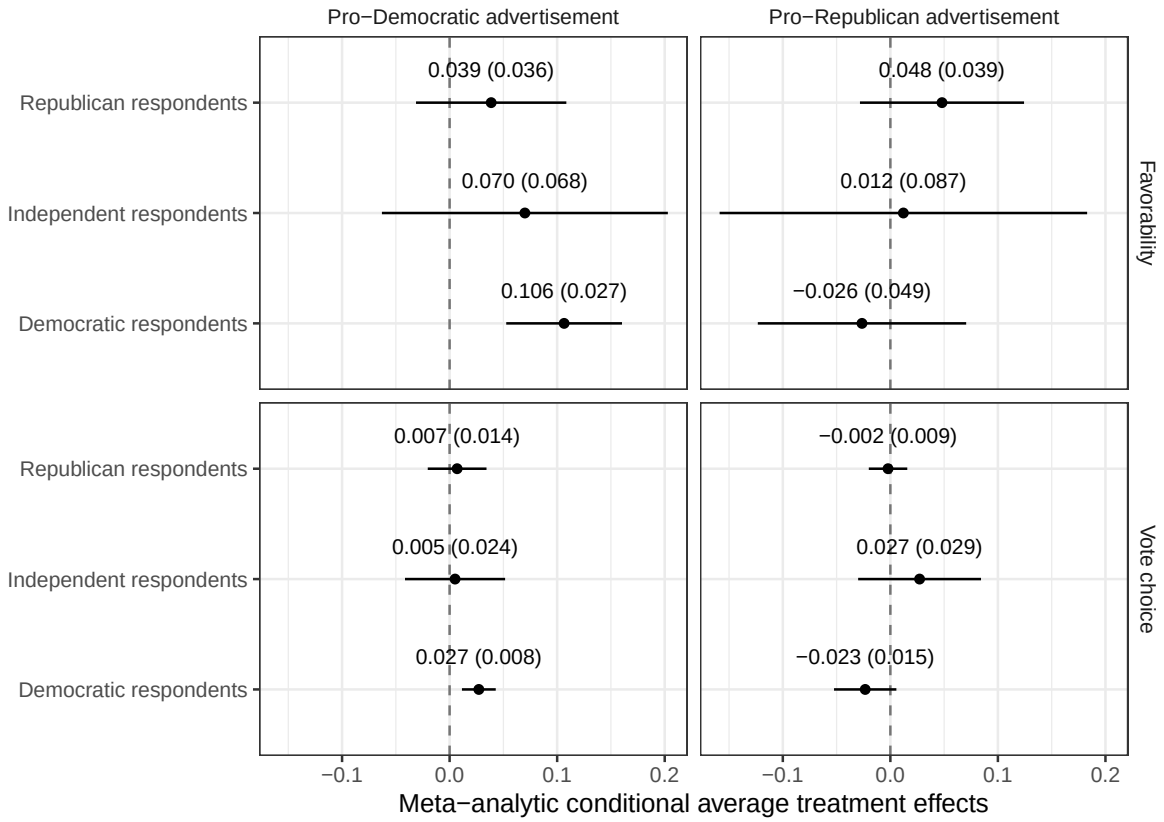


Figure 2: Figure 2 as reproduced by the maintained rewrite.

Figure S1's seven panels carry no printed numbers. Its seventeen estimates agree with the archive's to within 3e-17, and the panel set is derived from the data rather than typed in: `figure_s1_repeat_ads.R` asserts that the seven advertisements it plots are exactly the advertisements the data show in more than one week.

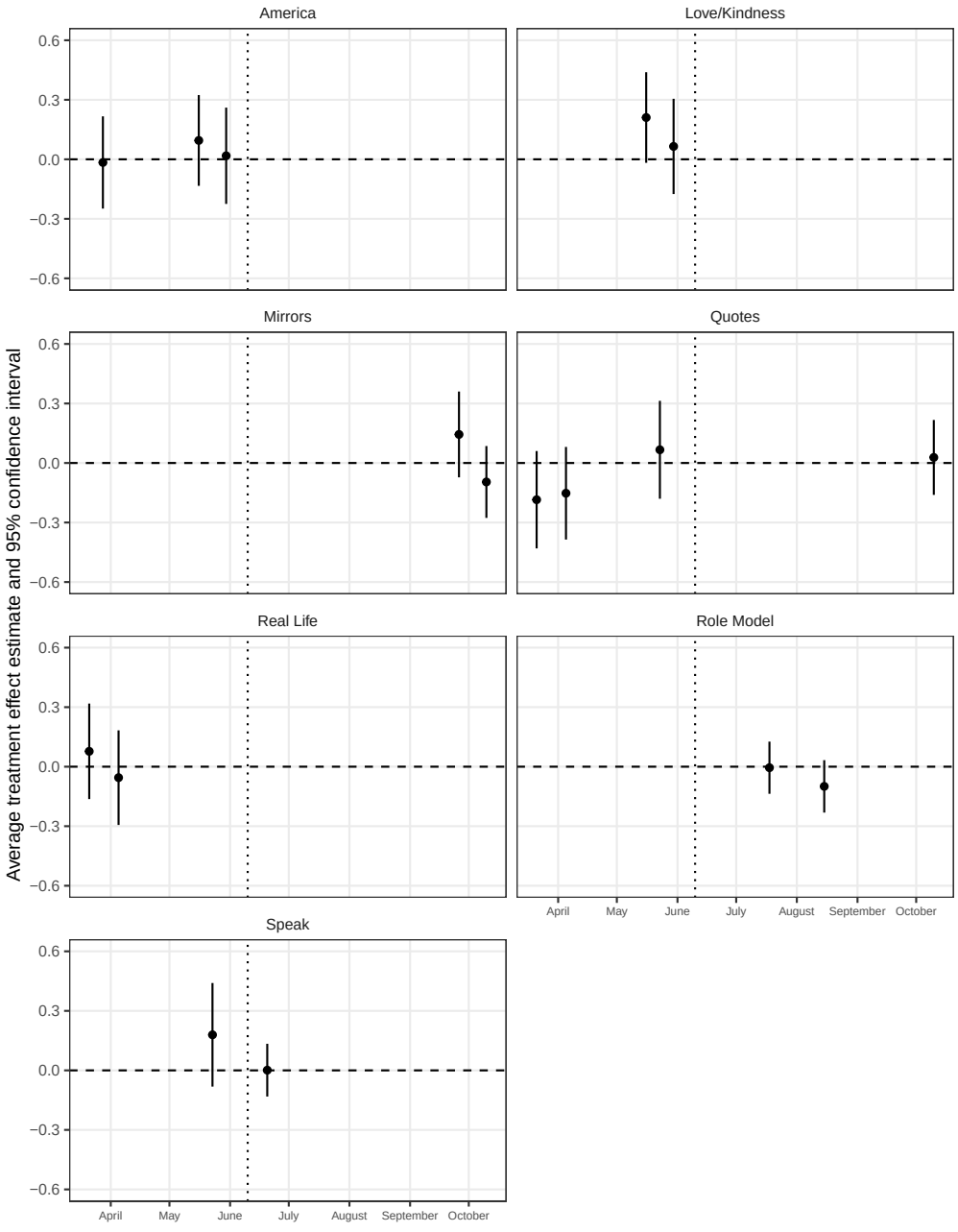


Figure 3: Figure S1 as reproduced by the maintained rewrite.

9 Maintained rewrite verification

Table 8: Maintained rewrite verification: published value against rewrite output.

Location	Quantity	Paper	Rewrite	Match	Locus
Title	Advertisement-week experiments	59	59	1	
Abstract	Distinct campaign advertisements tested	49	49	1	
Abstract	Advertisement-week experiments	59	59	1	
Abstract	Respondents completing a survey	34000	34000	1	
Introduction	Months spanned by the fielding period	8	7.81930184804928	1	
Introduction	Distinct campaign advertisements tested	49	49	1	
Introduction	Distinct campaign advertisements, restated	49	49	1	
Introduction	Pooled favorability effect (scale points)	0.05	0.0492278297226765	1	
Introduction	SD of favorability effects across experiments (scale points)	0.07	0.0681856141500513	1	
Introduction	Pooled vote choice effect (percentage points)	0.7	0.718114897405216	1	
Introduction	SD of vote choice effects (percentage points)	2	2.2186653517172	1	
Introduction	Weekly estimates exclude large persuasive effects				
Materials and Methods	Weeks of fielding	29	29	1	
Materials and Methods	Smallest weekly sample	1000	1000	1	
Materials and Methods	Largest weekly sample	2000	2000	1	
Materials and Methods	Share starting but not finishing (%)	42	42.4450688966381	1	
Materials and Methods	Attrition tests conducted	29	29	1	
Materials and Methods	Attrition tests significant unadjusted	3	3	1	
Materials and Methods	Attrition tests significant after Holm	0	0	1	
Materials and Methods	Attrition tests significant after Benjamini-Hochberg	0	0	1	
Materials and Methods	Anti-Trump advertisement deployments	30	30	1	
Materials and Methods	Anti-Clinton advertisement deployments	11	11	1	
Materials and Methods	Pro-Clinton advertisement deployments	8	8	1	
Materials and Methods	Pro-Trump advertisement deployments	3	3	1	
Materials and Methods	Control group manipulation check, minimum (%)	2	1.5625	1	
Materials and Methods	Control group manipulation check, maximum (%)	4	4.01459854014598	1	
Materials and Methods	Treatment group manipulation check, minimum (%)	92	91.4156626506024	0	paper_internal
Materials and Methods	Treatment group manipulation check, maximum (%)	95	95.5431754874652	0	paper_internal
Materials and Methods	Respondents in a single weekly study	1000	1000	1	
Results	Experimental comparisons plotted in Figure 1	59	59	1	
Results	Pooled favorability effect (scale points)	0.049	0.0492278297226765	1	
Results	The pooled favorability effect is statistically significant		1	1	
Results	Pooled vote choice effect (percentage points)	0.7	0.718114897405216	1	
Results	The pooled vote choice effect is not statistically significant		1	1	
Results	Favorability SATE homogeneity p-value	0.09	0.0919044204718359	1	
Results	Vote choice SATE homogeneity p-value	0.07	0.0699520982406853	1	
Results	Largest significant negative favorability estimate in May	-0.5	-0.520943603069464	1	
Results	Largest significant positive favorability estimate in October	0.5	0.43202600389018	0	paper_internal
Results	Other significant negative favorability estimates in October	2	2	1	
Results	Smallest weekly sample	1000	1000	1	
Results	Largest weekly sample	2000	2000	1	
Results	Partisanship cells of Table 1 whose standard error exceeds the coefficient		6	0	paper_internal
Results	No partisanship coefficient in Table 1 is distinguishable from zero		0	1	
Results	Specifications in which the time slope is negative	3	3	1	
Results	Specifications in which the time slope is distinguishable from zero	1	1	1	
Results	Battleground difference in favorability, magnitude	0.008	0.0076808718197986	1	
Results	General election coefficient (scale points)	0.121	0.122624298396105	0	paper_internal
Results	The general election coefficient is not statistically significant		1	1	
Results	Upper bound of the general election interval	0.25	0.253750226781251	1	
Results	No tone or sponsor coefficient in Table 1 is distinguishable from zero		0	1	

Table 8: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Quantity	Paper	Rewrite	Match	Locus
Results	Target candidate coefficients relative to the omitted pro-Clinton category		6		
Results	No target candidate coefficient in Table 1 is distinguishable from zero		0	1	
Results	Democratic respondents respond more strongly to pro-Democratic than pro-Republican advertisements		1	1	
Results	Republican respondents respond similarly to pro-Democratic and pro-Republican advertisements				
Results	SATE favorability homogeneity p-value, restated	0.09	0.0919044204718359	1	
Results	SATE vote choice homogeneity p-value, restated	0.07	0.0699520982406853	1	
Results	CATE favorability homogeneity p-value	0.0002	0.00016505690635 5207	1	
Results	CATE vote choice homogeneity p-value	0.96	0.0961039371758417	0	paper_internal
Results	Homogeneity tests in Table S2 that fail to reject at the 5 per cent level	3	3	1	
Results	Tau, favorability SATEs	0.07	0.0681856141500513	1	
Results	Tau, favorability CATEs	0.15	0.147422967808586	1	
Results	Tau, vote choice SATEs	0.02	0.022186653517172	1	
Results	Tau, vote choice CATEs	0.02	0.0232088109488414	1	
Discussion	Advertisement-week experiments	59	59	1	
Discussion	Respondents completing a survey	34000	34000	1	
Discussion	Distinct campaign advertisements tested	49	49	1	
Figure 1	Favorability meta-analytic estimate	0.049	0.0492278297226765	1	
Figure 1	Favorability meta-analytic standard error	0.020	0.0200009449806686	1	
Figure 1	Vote choice meta-analytic estimate	0.007	0.00718114897405216	1	
Figure 1	Vote choice meta-analytic standard error	0.007	0.00732724536941071	1	
Table 1	Table 1, model 1, intrcpt, estimate	0.056	0.0559513746296691	1	
Table 1	Table 1, model 1, intrcpt, std.error	0.020	0.0203091094511682	1	
Table 1	Table 1, model 2, intrcpt, estimate	0.062	0.0619557871802225	1	
Table 1	Table 1, model 2, intrcpt, std.error	0.020	0.0204550273103208	1	
Table 1	Table 1, model 3, intrcpt, estimate	0.007	0.0071734391731233	1	
Table 1	Table 1, model 3, intrcpt, std.error	0.007	0.0071442025839593	1	
Table 1	Table 1, model 4, intrcpt, estimate	0.008	0.00761340879292042	1	
Table 1	Table 1, model 4, intrcpt, std.error	0.007	0.00721776470356415	1	
Table 1	Table 1, model 1, democrat_d, estimate	0.035	0.035239216310936	1	
Table 1	Table 1, model 1, democrat_d, std.error	0.035	0.0346822419625252	1	
Table 1	Table 1, model 2, democrat_d, estimate	0.022	0.0217616219401439	1	
Table 1	Table 1, model 2, democrat_d, std.error	0.036	0.0361069235707198	1	
Table 1	Table 1, model 3, democrat_d, estimate	0.011	0.010806619726571	1	
Table 1	Table 1, model 3, democrat_d, std.error	0.010	0.010032366314141	1	
Table 1	Table 1, model 4, democrat_d, estimate	0.006	0.00607624424823846	1	
Table 1	Table 1, model 4, democrat_d, std.error	0.011	0.0111901459055996	1	
Table 1	Table 1, model 1, independent_d, estimate	0.023	0.0230994168403524	1	
Table 1	Table 1, model 1, independent_d, std.error	0.051	0.0514908172832447	1	
Table 1	Table 1, model 2, independent_d, estimate	0.015	0.0148795264572918	1	
Table 1	Table 1, model 2, independent_d, std.error	0.052	0.0518214653308673	1	
Table 1	Table 1, model 3, independent_d, estimate	0.009	0.00882756839601492	1	
Table 1	Table 1, model 3, independent_d, std.error	0.020	0.0196989007331031	1	
Table 1	Table 1, model 4, independent_d, estimate	0.007	0.00662959611432708	1	
Table 1	Table 1, model 4, independent_d, std.error	0.020	0.0199216048048094	1	
Table 1	Table 1, model 1, battleground_d, estimate	-0.00	-0.0076808718197986	0	paper_internal
Table 1	Table 1, model 1, battleground_d, std.error	0.033	0.0330993309997604	1	
Table 1	Table 1, model 2, battleground_d, estimate	-0.007	-0.00728897411725834	1	
Table 1	Table 1, model 2, battleground_d, std.error	0.033	0.0331245308366241	1	
Table 1	Table 1, model 3, battleground_d, estimate	-0.017	-0.0169737781355814	1	
Table 1	Table 1, model 3, battleground_d, std.error	0.010	0.00982144227142084	1	
Table 1	Table 1, model 4, battleground_d, estimate	-0.017	-0.0170054042184294	1	
Table 1	Table 1, model 4, battleground_d, std.error	0.010	0.00996790251931272	1	
Table 1	Table 1, model 1, pac_d, estimate	-0.012	-0.0124947275778723	1	
Table 1	Table 1, model 1, pac_d, std.error	0.043	0.0426308784476082	1	
Table 1	Table 1, model 2, pac_d, estimate	0.026	0.0256893089604678	1	
Table 1	Table 1, model 2, pac_d, std.error	0.047	0.0469851815197419	1	
Table 1	Table 1, model 3, pac_d, estimate	-0.023	-0.0231494095931858	1	
Table 1	Table 1, model 3, pac_d, std.error	0.013	0.0128349559721876	1	
Table 1	Table 1, model 4, pac_d, estimate	-0.016	-0.0156649998403321	1	
Table 1	Table 1, model 4, pac_d, std.error	0.014	0.0144268706850987	1	

Table 8: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Quantity	Paper	Rewrite	Match	Locus
Table 1	Table 1, model 1, date_d, estimate	-0.023	-0.0230094166284181	1	
Table 1	Table 1, model 1, date_d, std.error	0.014	0.0140563401835011	1	
Table 1	Table 1, model 2, date_d, estimate	0.005	0.0045562321145179	1	
Table 1	Table 1, model 2, date_d, std.error	0.010	0.0101010468834039	1	
Table 1	Table 1, model 3, date_d, estimate	-0.009	-0.00937763297831397	1	
Table 1	Table 1, model 3, date_d, std.error	0.004	0.00379382342908776	1	
Table 1	Table 1, model 4, date_d, estimate	-0.008	-0.00774541926272452	1	
Table 1	Table 1, model 4, date_d, std.error	0.004	0.00426985776830753	1	
Table 1	Table 1, model 1, attack_d, estimate	-0.017	-0.0171013743308074	1	
Table 1	Table 1, model 1, attack_d, std.error	0.046	0.0462370739155849	1	
Table 1	Table 1, model 3, attack_d, estimate	0.028	0.0279600502401837	1	
Table 1	Table 1, model 3, attack_d, std.error	0.016	0.0163558547902618	1	
Table 1	Table 1, model 1, general_d, estimate	0.123	0.122624298396105	1	
Table 1	Table 1, model 1, general_d, std.error	0.067	0.0669022132138403	1	
Table 1	Table 1, model 2, pro_trump_d, estimate	-0.124	-0.123836901513909	1	
Table 1	Table 1, model 2, pro_trump_d, std.error	0.101	0.101118675428876	1	
Table 1	Table 1, model 4, pro_trump_d, estimate	-0.016	-0.0157727126722738	1	
Table 1	Table 1, model 4, pro_trump_d, std.error	0.034	0.0337767139387826	1	
Table 1	Table 1, model 2, anti_clinton_d, estimate	-0.105	-0.105067385465675	1	
Table 1	Table 1, model 2, anti_clinton_d, std.error	0.070	0.0698886090607599	1	
Table 1	Table 1, model 4, anti_clinton_d, estimate	0.012	0.0115687385179081	1	
Table 1	Table 1, model 4, anti_clinton_d, std.error	0.023	0.02296757258421	1	
Table 1	Table 1, model 2, anti_trump_d, estimate	-0.041	-0.0413676950109048	1	
Table 1	Table 1, model 2, anti_trump_d, std.error	0.058	0.0579066875305163	1	
Table 1	Table 1, model 4, anti_trump_d, estimate	0.026	0.0261014333861311	1	
Table 1	Table 1, model 4, anti_trump_d, std.error	0.021	0.0214091583815356	1	
Table 1	Table 1, model 2, pro_sanders_d, estimate	-0.075	-0.0749244096818968	1	
Table 1	Table 1, model 2, pro_sanders_d, std.error	0.089	0.0885300550768095	1	
Table 1	Table 1, model 2, pro_cruz_d, estimate	0.047	0.0470694919223399	1	
Table 1	Table 1, model 2, pro_cruz_d, std.error	0.116	0.116193805568345	1	
Table 1	Table 1, model 2, pro_kasich_d, estimate	-0.182	-0.181777858909213	1	
Table 1	Table 1, model 2, pro_kasich_d, std.error	0.145	0.14476493785976	1	
Table 1	Table 1, model 1, number of observations	354	354	1	
Table 1	Table 1, model 2, number of observations	354	354	1	
Table 1	Table 1, model 3, number of observations	204	204	1	
Table 1	Table 1, model 4, number of observations	204	204	1	
Table 1	Table 1 marks the model 1 average effect as significant at the 5 per cent level		1	1	
Table 1	Table 1 marks the model 2 average effect as significant at the 5 per cent level		1	1	
Table 1	Table 1 marks the model 3 time slope as significant at the 5 per cent level		1	1	
Figure 2	Figure 2, Favorability, Republican respondents, pro-Democratic ad, estimate	0.039	0.0386595396179065	1	
Figure 2	Figure 2, Favorability, Republican respondents, pro-Democratic ad, std.error	0.036	0.0356581601671249	1	
Figure 2	Figure 2, Favorability, Republican respondents, pro-Republican ad, estimate	0.048	0.0480196665227286	1	
Figure 2	Figure 2, Favorability, Republican respondents, pro-Republican ad, std.error	0.039	0.0389230315207782	1	
Figure 2	Figure 2, Favorability, Independent respondents, pro-Democratic ad, estimate	0.070	0.0699932736594406	1	
Figure 2	Figure 2, Favorability, Independent respondents, pro-Democratic ad, std.error	0.068	0.0678203342309325	1	
Figure 2	Figure 2, Favorability, Independent respondents, pro-Republican ad, estimate	0.012	0.0121099394831964	1	
Figure 2	Figure 2, Favorability, Independent respondents, pro-Republican ad, std.error	0.087	0.0871736863408394	1	
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Democratic ad, estimate	0.106	0.106476573620707	1	
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Democratic ad, std.error	0.027	0.0274578187061506	1	
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Republican ad, estimate	-0.026	-0.0263833309845451	1	
Figure 2	Figure 2, Favorability, Democratic respondents, pro-Republican ad, std.error	0.049	0.0494319844445353	1	
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Democratic ad, estimate	0.007	0.00699092320146246	1	
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Democratic ad, std.error	0.014	0.0139356630863744	1	

Table 8: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Quantity	Paper	Rewrite	Match	Locus
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Republican ad, estimate	-0.002	-0.00216608514888707	1	
Figure 2	Figure 2, Vote choice, Republican respondents, pro-Republican ad, std.error	0.009	0.00912830313876472	1	
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Democratic ad, estimate	0.005	0.00506265749355459	1	
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Democratic ad, std.error	0.024	0.0237475886684295	1	
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Republican ad, estimate	0.027	0.0271627314372247	1	
Figure 2	Figure 2, Vote choice, Independent respondents, pro-Republican ad, std.error	0.029	0.0291458503455212	1	
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Democratic ad, estimate	0.027	0.0271759277742814	1	
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Democratic ad, std.error	0.008	0.00801715594366605	1	
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Republican ad, estimate	-0.023	-0.023396092643628	1	
Figure 2	Figure 2, Vote choice, Democratic respondents, pro-Republican ad, std.error	0.015	0.0147860283788054	1	
Appendix A	Distinct campaign advertisements	50	49	0	paper_internal
Appendix A	Advertisements fielded in more than one week	8	7	0	paper_internal
Appendix A	Anti-Trump deployments	28	30	0	paper_internal
Appendix A	Anti-Clinton deployments	9	11	0	paper_internal
Appendix A	Pro-Clinton deployments	9	8	0	paper_internal
Appendix A	Pro-Trump deployments	3	3	1	
Table S1	Weeks listed	29	29	1	
Table S1	Advertisement-week entries	58	58	1	
Appendix B	Advertisements fielded more than once	7	7	1	
Appendix B	Repeat advertisements whose effect varies significantly across weeks		0	1	
Appendix B	Weekly estimates for the advertisement Quotes				
Figure S1	Panels	7	7	1	
Figure S1	Weekly estimates plotted	17	17	1	
Table S2	Table S2, sates, Favorability, estimate	0.0492	0.0492278297226765	1	
Table S2	Table S2, sates, Favorability, std.error	0.0200	0.0200009449806686	1	
Table S2	Table S2, sates, Favorability, p.value	0.0919	0.0919044204718359	1	
Table S2	Table S2, sates, Favorability, tau2	0.0046	0.00464927797701968	1	
Table S2	Table S2, sates, Favorability, tau	0.0682	0.0681856141500513	1	
Table S2	Table S2, sates, Vote Choice, estimate	0.0072	0.00718114897405216	1	
Table S2	Table S2, sates, Vote Choice, std.error	0.0073	0.00732724536941071	1	
Table S2	Table S2, sates, Vote Choice, p.value	0.0700	0.0699520982406853	1	
Table S2	Table S2, sates, Vote Choice, tau2	0.0005	0.000492247594291039	1	
Table S2	Table S2, sates, Vote Choice, tau	0.0222	0.022186653517172	1	
Table S2	Table S2, cates, Favorability, estimate	0.0585	0.0584522714979407	1	
Table S2	Table S2, cates, Favorability, std.error	0.0178	0.0178394276247739	1	
Table S2	Table S2, cates, Favorability, p.value	0.0002	0.000165056906355207	1	
Table S2	Table S2, cates, Favorability, tau2	0.0217	0.0217335314374913	1	
Table S2	Table S2, cates, Favorability, tau	0.1474	0.147422967808586	1	
Table S2	Table S2, cates, Vote Choice, estimate	0.0083	0.00826356365738537	1	
Table S2	Table S2, cates, Vote Choice, std.error	0.0051	0.00514489865140912	1	
Table S2	Table S2, cates, Vote Choice, p.value	0.0961	0.0961039371758417	1	
Table S2	Table S2, cates, Vote Choice, tau2	0.0005	0.000538648905659061	1	
Table S2	Table S2, cates, Vote Choice, tau	0.0232	0.0232088109488414	1	

326 of 342 published quantities are reproduced by the maintained rewrite to the precision the article reports. The 12 that are not are the errata above, where the archive and the rewrite agree with each other and the article says something else.

No published number appears anywhere in `maintained/` as an input to a computation. Every value in `maintained/output/` traces to `original/exps.rds`, `original/exps_with_incompletes.rds` or `original/ad_df.csv`. Published values appear only in `ground_truth/build_ground_truth.R`,

where they are comparison targets and nothing reads them back.

Running `run_all.R` twice returns every CSV, TeX, PNG and RDS file in `maintained/output/` byte-identical. Only the three figure PDFs differ, because a PDF records the time it was written.

10 R environment

Item	Value
R version	4.6.2
Platform	aarch64-apple-darwin23
Date run	2026-11-13

Table 10: Package versions used for the run behind this report.

Package	Version
estimatr	2.0.1
metafor	5.2.1
modelsummary	2.6.0
janitor	2.2.1
broom	1.0.13
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
readr	2.2.0
here	1.0.2